

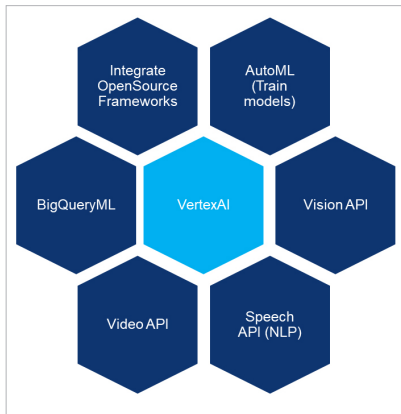


Figure 3: Vertex AI APIs

do advanced analytics to get predictive results like forecasting bill details for the next quarter based on the invoices fed into the parser for previous quarters.

Google Vertex AI

Google Cloud Platform (GCP) has announced an integrated managed ML platform called Vertex AI. This is a unified platform that combines all the ML services of GCP in a single pane including AutoML for training and developing ML models, vision API for image processing; speech API for translation, speech-to-text and vice-versa; video API for streaming videos, and BigQueryML to build data to the AI life cycle for business intelligence processing. It also supports open source platform integration like TensorFlow, scikit-learn and PyTorch to create a complete AI toolkit.

Vertex AI has multiple sub-components like Vertex model monitoring, Vertex ML metadata and Vertex pipeline for end-to-end ML workflow activities including monitoring, building/deployment and tracking/querying of ML models. It has a unified GUI for the dashboard, ML model building, training the model and deploying the services. Any custom coding for MLOps or ML model development can be reduced by up to 80 per cent.

The Vertex Feature Store can be used to build ML models and share,

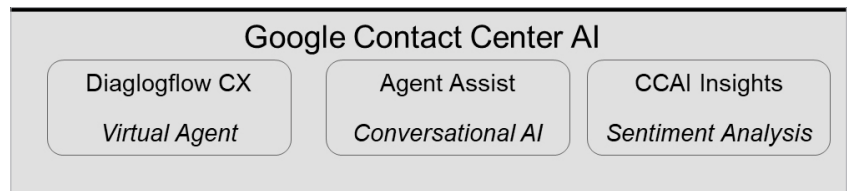


Figure 4: Google Contact Center AI

serve and use ML features in the Marketplace. The Vertex AI platform also has prebuilt deep learning VM images so that we can quickly spin-up these instances to develop applications using popular AI frameworks like TensorFlow or PyTorch (similar to Jupyter notebooks).

Google Contact Center AI

The pandemic has made customers hesitant to visit banks or retail stores — they prefer to use chatbots or virtual assistants to interact with them. Google Cloud provides loads of AI based solutions that are integrated with Google Contact Center AI services for virtual assistance.

Some of the important components of Google's Contact Center AI (CCAI) are briefly described below.

Dialogflow CX: During customer assistance calls, complex queries and supplementary questions need to be answered. This means the required responses need to be elicited from the knowledge base to frame appropriate replies to the customer. Dialogflow CX agents use Google's natural language understanding (NLU) algorithms to handle these search capabilities.

Agent Assistant: This component is built with AI assisted conversational services to prepare recommended answers for human agents to choose from while replying to customer queries. It uses a centralised knowledge base to prepare ready-to-use responses and also transcribe live calls in different languages. This enables processing in the backend for training AI models to analyse frequently asked questions, answer patterns and customer groups based on interest.

CCAI Insights: From voice enabled calls and chat based text analysis, CCAI Insights helps to prepare metrics analysis, sentiment analysis and ratings for answers. It uses the NLP capabilities of Google Cloud to learn about customer interactions and improve the connect with them (chat/voice).

Google CCAI services include text-to-speech and speech-to-text conversion, sentiment analysis, ranking of responses, integrated IVR services and NLP to provide unified customer support.

Google Natural Language AI

Classifying text and processing it for inference is very important in the case of AI based solutions. Medical analytics from unstructured medical data, news analysis from unstructured news feeds, and marketing analysis from retail data are common use cases.

Google Cloud Platform (GCP) provides a Natural Language API for such use cases. This API helps to extract, analyse and store text from unstructured and unclassified data sets by applying natural language understanding (NLU) using services like speech-to-text API, vision API and translator API.

Speech-to-text API is used to convert audio to text and extract meaningful insights for any text processing activities. As an example, you can use this API for extracting data, processing it and gaining insights from a doctor's voice note to prepare a medical prescription.

Similarly, optical character recognition (OCR) can be achieved using the vision API. Cheque processing in banks, handwritten character recognition for application form processing in hospitals and banks, and insurance claim processing from submitted bills are a few use cases of this API.